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SHORT PAPER

Slip Ratio Prediction in Autonomous Wheeled Robot using ROS-Physics Engine based Hybrid Classification Approaches

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Abstract
This study presents an approach for predicting the slip ratio, a significant parameter in improving the autonomous navigation of robots. The novelty of the work is presented in two parts; first is the design of a Robot Operating System (ROS)-Physics Engine-based hybrid machine learning model with separate terrain classification and slip ratio estimation algorithms. The motivation for developing a hybrid model is presented by analyzing slip ratio-longitudinal friction coefficient characteristics in three different terrains: sand, grass, and asphalt. The longitudinal friction coefficient (μ_x) values are distinct for the same values of slip ratio in different terrains, potentially influencing the model's feature extraction. The features for the slip estimation model are collected from torque sensors and IMU, which measure the external forces acting on robot wheels. The slip estimation is achieved using ensembles of weak classifiers to reduce the computational load of onboard systems. An optimized model with slip ratio classes as output has been evaluated and presented for three different terrains. The model provides an accuracy of 78.1% in the sand, 77.8% in the grass, and 73.5% in asphalt in simulations. The model is compared with a strong classifier used in previous works regarding the accuracy, training time, and computational time. Second, the virtually- trained learning algorithm is tested and validated using the physical robot whose virtual model is implemented in simulations with the actual slip ratio estimated using an experimental framework.

Keywords Mobile robot · Machine learning · Transfer learning · Terrain parameters · Slip ratio

1 Introduction

ROBOTS are inherently intelligent devices that can act and sense according to their surroundings. The growth of mobile robots is invigorated by the need for such devices for arduous tasks like planetary explorations, defence missions, rescue purposes, and different applications related to Industry 4.0. The development of machine learning methods and advanced control algorithms steer the development of intelligent robotic systems. This work is motivated by the need for data-driven systems in robust control design for mobile robots. Unlike conventional control approaches, data-driven mechanisms provide rational perception, critical in state estimation and feedback to the popular strategies like Model Predictive Control (MPC). The navigation of mobile robots is challenging in different dimensions, such as localization, path and motion planning, obstacle avoidance to name a few [1–3].

Along with these, the physical properties of the navigation surface influence the robot's locomotion. For autonomous driving, the robot needs to sense and act to the terrain, just like in the case of manual driving, where the driver changes the vehicle's speed or even avoids the path in response to the road condition. These studies can be of three dimensions; terrain identification, classification, and parameter estimation. A good perception of these three is necessary from the control perspective for smooth navigation. The development of machine learning methods and the possibilities of integrating perception skills with robots made the research area more attractive.

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